

The consolidated U-series verification script: one draft script covering every machine check registered in units U2 through U12

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Units U2–U12 registered eleven machine checks across three claims; none could be executed in this session (no shell). This unit converts the registry into one implementable artifact: a single draft script with one section per check, faithful to the production conventions (the same m424 helpers and J -integral form as the certified 192/192 suite). Scope: tooling only — no mathematical content beyond the notes’ own arithmetic, now encoded as asserts.

2. Check coverage map

Section	Registered check (source unit)	Key asserts
S1	G-A0-VER (U2)	$M_c = 43/1620$ exact; $m_w = 0.0392414$; gap-identity
S2	M-ENDPOINT (U7)	$m^* = r_R$ to 2×10^{-5} ; window $\times 7+$; $M_R/M_c > 4$
S3	U4/U7 tables	$M(\hat{r}(I)) < M_R$ at all anchors; endpoint value inside the U7 bracket with convexity allowance
S4	R-U6-2 (U6)	$J(0) = 0.290 \pm 0.01$; composed margins;
S5	R-U10-3 (U10/U11)	frozen-coupling ratios 7.7/2.8/0.97 (8% tol); dressed ratios $\sim 3.2/ \sim 1.34$ (15% tol); endpoint positivity
S6	U8 angle table	$3u_{\text{eff}}M_R = 3uM_R + 30vM_R^2$ exact (10^{-12})
S7	U12 ceiling	$-M'(\hat{r}) = J(0)$ to 2% (finite difference); linear gain $2\lambda'M_R = 1.763$; endpoint convention remainder $\leq 3.3 \times 10^{-5}$; protection/remainder > 100
		FCC 70.5288° , icosahedral 63.4349° exact; $\theta_{\min}$ anchors; all eight patterns in-class at all I ; decagonal extremal margin $< 1\%$ at the endpoint
		$\log_2 88 = 6.459$; sharpening factor 2.54; monotonicity

Not covered (deliberately, larger scope): the RES-4 interval sweep (its own driver), the U13 χ spot-check (needs the main script’s configuration generators — registered as an extension of that suite), the R-U10-3 trace-inequality test on full patterns (needs the Bloch machinery — extension of the main suite), the U12 section-3 definition check (reads the main script’s class conventions — extension there).

3. Execution contract

python codes/vacuum/t6_mainline_useries_checks.py — expected: all claims PASS, exit 0, runtime well under the 45 s cap (only M_fast/gap_solve calls plus two J quadratures); artefact

claims/B1-RH-ENUM/runs/260606-useries-checks/result.json. FIRST-RUN protocol: review every FAIL (the dressed-ratio tolerances encode estimate-grade numbers — a FAIL there demotes the corresponding U-note figure per its own pre-registered gate, it does NOT demote the structural lemmas); on full PASS, flip the script’s `__status__` and update the U-note grades per their footers.

4. Numerical verification

This unit IS the verification tooling; it introduces no new numbers beyond the notes’ own (every assert’s expected value cites its source unit in the coverage map). Quantitative sanity check (mandatory): self-consistency of the assert set — S4’s identity is algebraically exact (the two expressions are the same polynomial in M_R), so its 10^{-12} tolerance is a tautology-check of the script’s own arithmetic, deliberately included as the canary assert (if IT fails, the script’s environment, not the mathematics, is broken); the S5 finite-difference step $h = 10^{-5}$ sits four orders below the $\hat{r}$ scale and two above double-precision noise for `M_fast` — the standard sweet spot.

5. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “An unexecuted script can contain trivial bugs (typos, API drift) that make this unit’s value illusory.” VALID-with-mitigation: certain — which is why the script carries `__status__ = DRAFT - NOT YET EXECUTED` in its module docstring AND this note’s banner, the canary assert (S4) isolates environment breakage from mathematical failure, and the first-run protocol is written into section 3. The alternative (leaving eleven checks as prose registrations) is strictly worse for the next session.
- (β) “Encoding estimate-grade numbers as asserts with tolerances risks circularity: the script ‘verifies’ what the notes guessed.” DISMISSED with the design: the asserts’ EXPECTED values are the notes’ arithmetic, but the ACTUAL values come from the production helpers (`M_fast`, the J quadrature) — an independent computation path; agreement is verification, disagreement fires the notes’ own pre-registered gates. The dressed-ratio tolerances (15%) are wide precisely because those numbers are estimates pending M-ENDPOINT.
- (γ) “The script writes its artefact under B1/runs although it covers B2 and B5 checks too — claim-card hygiene.” VALID-with-mitigation: the single-artefact choice keeps the first run atomic; on full PASS the per-claim JSON split (B2’s S1, B5’s S3/S4/S7 slices) is a five-line follow-up, registered in the first-run protocol. The `__claims__` header lists all three claims explicitly.

6. Result footer

Result ID:	B1-RH-ENUM / useries-verification-script (unit U14)
Precise statement:	TOOLING: one draft script (codes/vacuum/t6_mainline_useries_checks.py v0.1.0, DRAFT - NOT YET EXECUTED) consolidating the eleven machine checks registered by units U2-U12, with per-check asserts, a canary identity, and a JSON artefact target.
Scope:	Tooling; no new mathematics; expected values

	cite their source units.
Dependencies:	m424 production helpers; the main suite's J-integral form; units U2-U12.
Evidence grade:	N/A (tooling) -- the script is DRAFT until first executed and reviewed.
Reproduction command:	python codes/vacuum/t6_mainline_useries_checks.py
Expected output:	all claims PASS, exit 0, < 20 s; artefact claims/B1-RH-ENUM/runs/260606-useries-checks/result.json. First-run protocol in section 3.
Falsification gate:	Any S-section FAIL fires the source unit's own pre-registered gate (table in section 2); an S4 canary FAIL indicates environment breakage, not mathematics.
Tier before / after:	No tier action anywhere.
No-overclaim statement:	NOTHING is machine-verified by authoring this script; every U-series number keeps its note's stated grade until the first reviewed run. PDF build + linter PENDING operator-side.
Next required action:	OPERATOR-SIDE (or next shell-capable session): execute the script, review, flip __status__, split per-claim artefacts, update U-note grades per their footers.